20261112 Quiz

for (MATH261) Discrete Mathematics

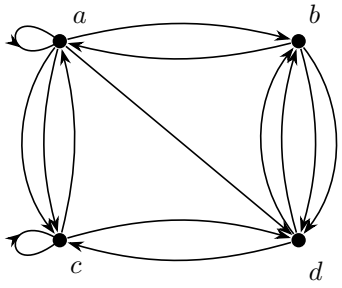
Jungwoo Kim
Department of Mathematics
Pohang University of Science and Technology

Problem 1.

(a) Draw the directed graph represented by the given adjacency matrix.

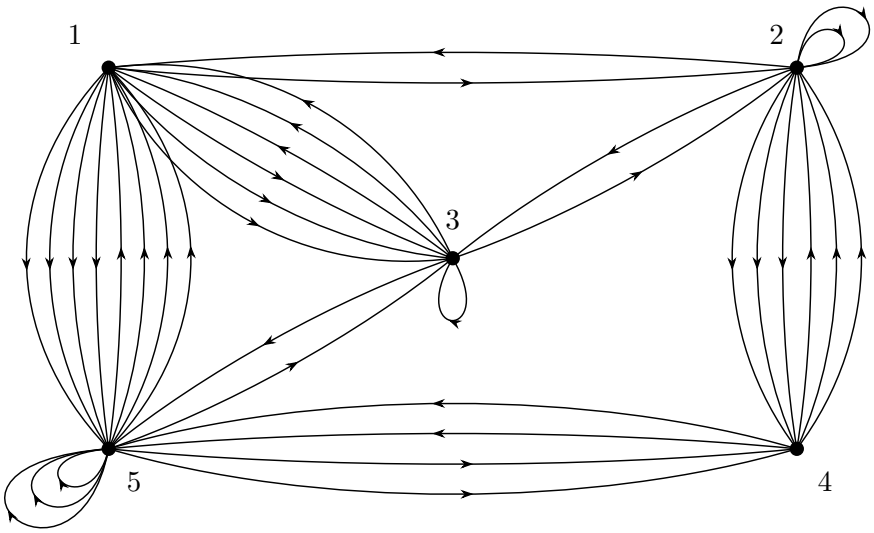
(b) Find the adjacency matrix of the given directed multigraph.

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 3 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 0 & 3 & 0 & 0 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{pmatrix}.$$



Solution.

(a) Drawing each arc separately, including the loops, gives the following directed multigraph.



(b) With rows and columns ordered as a, b, c, d, the adjacency matrix is

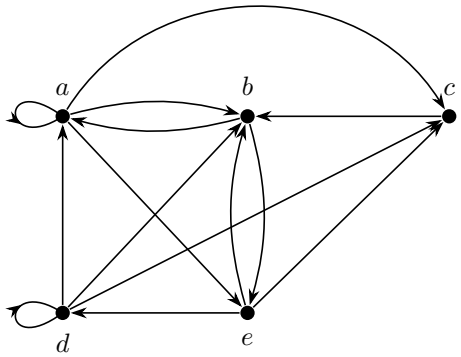
$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \end{pmatrix}.$$

□

Problem 2. Consider the following directed graph.

(a) Find all paths starting at vertex a and ending at vertex b or c. Determine the length of each path.

(b) Find all cycles starting at vertex a. Determine the length of each cycle.



Solution.

(a) A path cannot repeat a vertex. The paths are listed below, grouped by their terminal vertices; the number in parentheses is the length.

Ending at b	Ending at c
(a, b) (1)	(a, c) (1)
(a, c, b) (2)	(a, e, c) (2)
(a, e, b) (2)	(a, b, e, c) (3)
(a, e, c, b) (3)	(a, e, d, c) (3)
(a, e, d, b) (3)	(a, b, e, d, c) (4)
(a, e, d, c, b) (4)	

These 11 paths are exhaustive: the first edge from a goes to b, c, or e, and no vertex may be revisited.

(b) A cycle returns to its initial vertex without repeating any other vertex. The cycles starting at a, grouped by length, are

Length	Cycle
1	(a, a)
2	(a, b, a)
3	(a, c, b, a), (a, e, b, a), (a, e, d, a)
4	(a, b, e, d, a), (a, e, c, b, a), (a, e, d, b, a)
5	(a, c, b, e, d, a), (a, e, d, c, b, a)

There are 10 cycles in total, including the loop (a, a) and the directed 2-cycle (a, b, a) under the course definition. Checking the first edge from a and forbidding repeated intermediate vertices proves exhaustiveness.

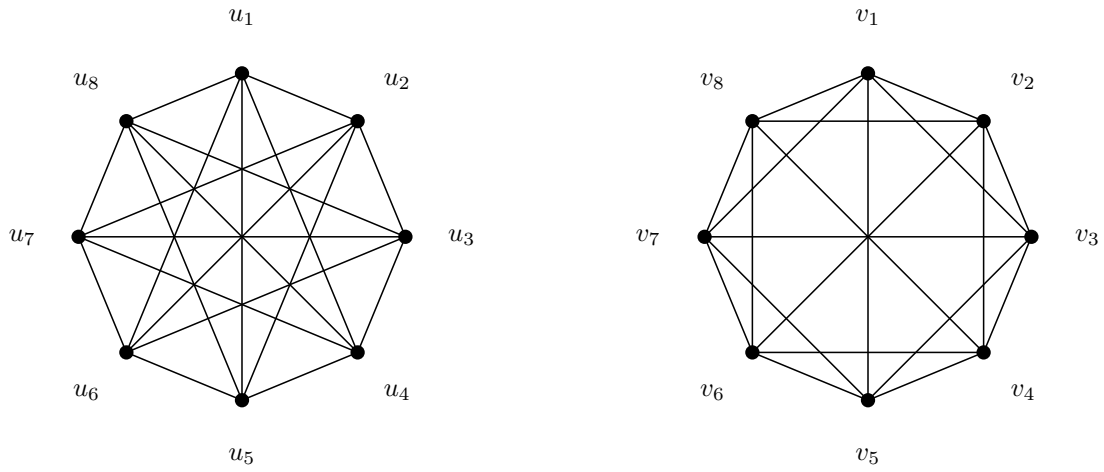
□

Problem 3. Show that isomorphism of simple graphs is an equivalence relation.

Solution. For every simple graph G, the identity on $V(G)$ is an isomorphism, so the relation is reflexive. If $f\colon V(G) \to V(H)$ is an isomorphism, then $f^{-1}\colon V(H) \to V(G)$ is also one, proving symmetry. Finally, if $f\colon V(G) \to V(H)$ and $g\colon V(H) \to V(K)$ are isomorphisms, so is $g \circ f$, proving transitivity. Thus graph isomorphism is an equivalence relation.

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Problem 4. Determine whether the given pair of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous argument that none exists.



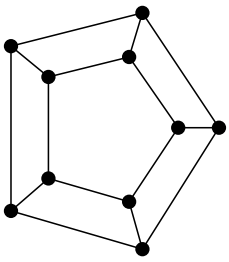
Solution. The graphs are *not* isomorphic. An isomorphism preserves both adjacency and nonadjacency, so it also induces an isomorphism between the complement graphs. Read the vertex subscripts modulo 8. In the left graph, u_i is adjacent to $u_{i\pm1}$, $u_{i\pm3}$, and u_{i+4}. Hence its complement consists of

$$u_1u_3u_5u_7u_1 \qquad \text{and} \qquad u_2u_4u_6u_8u_2,$$

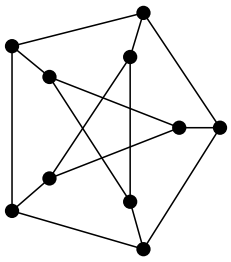
two disjoint 4-cycles. In the right graph, v_i is adjacent to $v_{i\pm1}$, $v_{i\pm2}$, and v_{i+4}, so its complement contains precisely the edges $v_iv_{i\pm3}$. Since $\gcd(3,8) = 1$, these form a single 8-cycle. The complements have different numbers of connected components and therefore cannot be isomorphic. It follows that the original graphs are not isomorphic.

□

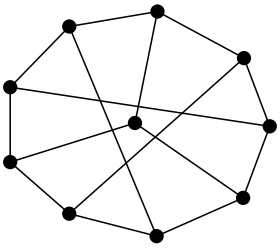
Problem 5. Determine which pairs of the following graphs are isomorphic. For each pair, either exhibit an isomorphism or provide a rigorous argument that none exists.



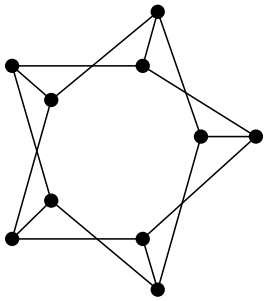
G_1



G_2



G_3



G_4

Solution. Only $G_2 \cong G_3$.

Step 1. We establish $G_2 \cong G_3$ by constructing an explicit isomorphism. Label the outer vertices of G_2 by x_0, x_1, x_2, x_3, x_4, clockwise from the top, and its corresponding inner vertices by y_0, y_1, y_2, y_3, y_4. In G_3, label the nine outer vertices by $z_0, \ldots, z_8$, clockwise from the top, and the central vertex by z_9. The following bijection preserves all 15 edges:

Vertex	x_0	x_1	x_2	x_3	x_4
Image	z_0	z_1	z_2	z_3	z_9
Vertex	y_0	y_1	y_2	y_3	y_4
Image	z_8	z_5	z_7	z_4	z_6

The outer 5-cycle maps to $z_0z_1z_2z_3z_9z_0$. The five spokes map to z_0z_8, z_1z_5, z_2z_7, z_3z_4, and z_9z_6; the inner edges map to z_8z_7, z_8z_4, z_5z_4, z_5z_6, and z_7z_6. These are exactly the edges of G_3, so $G_2 \cong G_3$.

Step 2. We show that G_1 is isomorphic to neither G_2 nor G_3. The graph G_1 contains a 4-cycle through its top outer, top inner, right inner, and rightmost outer vertices. The graph G_2 has no 4-cycle: its two 5-cycles are joined by a matching, so any 4-cycle would need two matching edges, one outer edge, and one inner edge. This would require the same two indices to be adjacent in both 5-cycles, which is impossible: outer edges join consecutive indices, while inner edges join indices differing by two. Since $G_2 \cong G_3$ and isomorphisms preserve cycle lengths, G_1 is isomorphic to neither.

Step 3. We show that G_4 is not isomorphic to any of the other three graphs. It consists of a 10-cycle together with five chords joining vertices at distance 5 along that cycle. Alternately coloring the cycle's vertices gives a bipartition, and each chord also joins opposite parts because 5 is odd. Thus G_4 is bipartite. By contrast, G_1, G_2, and G_3 each contain a 5-cycle and hence are not bipartite. Therefore G_4 is not isomorphic to any of them.

□